

Contextual Bandits

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Introduction: bandits with contexts

I. Linear Bandits:

I.1. LinUCB: Optimism in dimension d

I.2. LinTS: The Bayesian POV

II. Beyond Linear Models

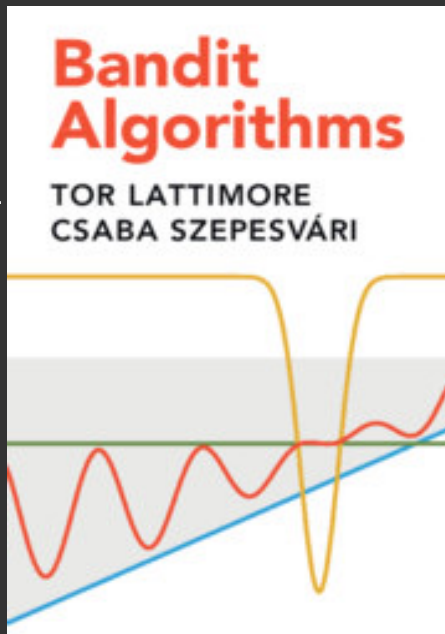
Conclusions and recap

Self Study

More on "contextual"
bandits :

Chap 18
(Exp 4)

+ connection to non-stationary
bandits : Chap 31

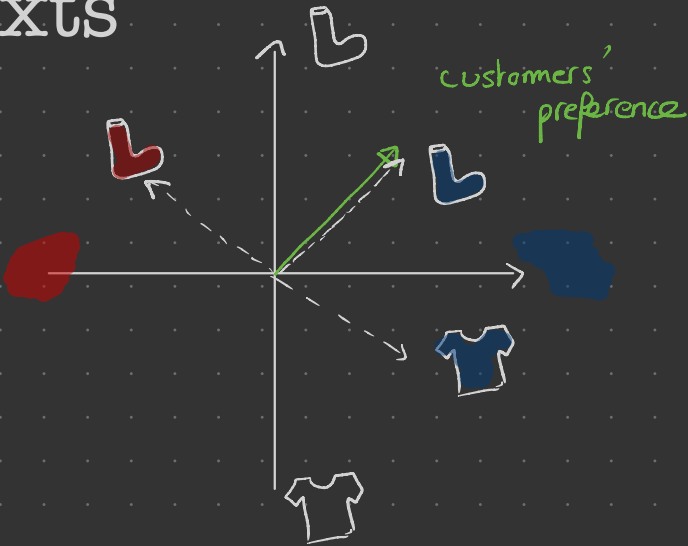


Today :

Chap. 19, 20, 21
(partially) + 36 (TS)

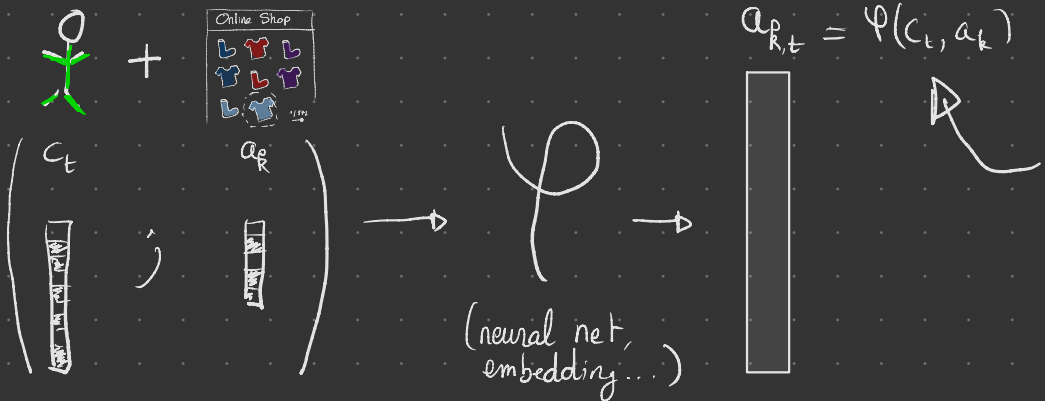
More on lower bounds:
Chap 24, 25

Introduction to Bandits with Contexts



I am attending RLSS in ? $\rightarrow \sqrt{1/x}$

Pipeline:



Contextualized action
vector in $\mathbb{R}^d$

Learning Setting

At time t ,

* A contextualized action set is revealed

$$\mathcal{A}_t = \{a_{t,1}, \dots, a_{t,k}\} \subset \mathbb{B}_{1,d}^k$$

* Learner chooses $A_t \in \mathcal{A}_t$

* Environment gives reward

$$R_t = f_\theta(A_t) + \eta_t$$

→ Parametrized reward function: θ encodes the similarity between actions.

→ Regret definition :

$$R(T) = \sum_{t=1}^T \max_{a \in \mathcal{A}_t} f_\theta(a) - f_\theta(A_t)$$

Goal: * Build an estimator $\hat{f}_\theta$ of f_θ

* Design an algorithm and show that it achieves sublinear regret

$$\mathbb{B}_{1,d} = \{x \in \mathbb{R}^d : \|x\| \leq 1\} \quad (\text{known})$$

I. Linear Models

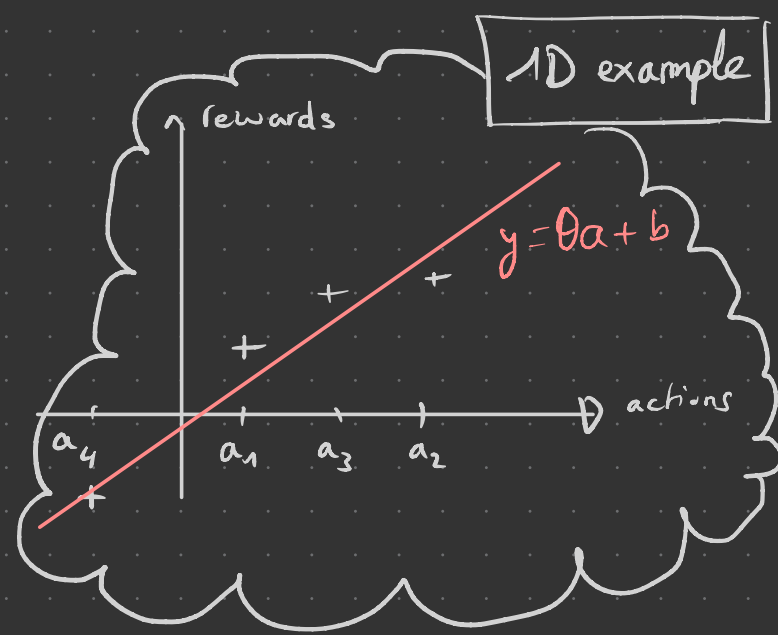
$$f_{\theta}(a) = a^{\top} \theta \rightarrow \text{“Stochastic Linear Bandits”}$$

Regularized least squares: Estimate $\hat{\theta}$

For each action $A_t \in \mathbb{R}^d$, we observe $Y_t = A_t^T \theta + \eta_t \in \mathbb{R}$

Assume $a_1, \dots, a_N$ are "fixed" ^{not random} vectors in $\mathbb{R}^d$ and $Y_1, \dots, Y_N$ the associated rewards.

→ How could you estimate θ ?



Regularized least squares

$$l(\theta; (a_i, Y_i)_{i=1}^N) = \sum_{i=1}^N (Y_i - \theta^T a_i)^2 + \lambda \|\theta\|_2^2 \quad \rightsquigarrow \quad \hat{\theta}^{\text{ALS}} = \underset{\theta}{\operatorname{argmin}} l(\theta; (a_i, Y_i)_{i=1}^N)$$

$$\nabla_{\theta} l = \sum_{i=1}^N \cancel{2} (Y_i - \theta^T a_i) a_i^T + \cancel{2} \lambda \theta^T$$

$$\nabla_{\theta} l = 0 \Leftrightarrow \sum Y_i a_i^T - \theta^T \left(\sum a_i a_i^T + \lambda I \right) = 0$$

$$\hat{\theta}^{\text{RLS}} = \underbrace{\left(\sum_{i=1}^N a_i a_i^T + \lambda I \right)^{-1}}_{\text{inverse covariance matrix}} \underbrace{\left(\sum_{i=1}^N Y_i a_i \right)}$$

$$\sum_i a_i \boxed{\boxed{a_i}}_d \rightarrow$$

"inverse covariance matrix"

Elliptical Uncertainty

Regularized least squares yield $\hat{\theta}_t^{(RLS)} = \underbrace{\left(\sum_{s=1}^{t-1} A_s A_s^T + \lambda I \right)^{-1}}_{V_t^{-1}(\lambda)} \left(\sum_{s=1}^{t-1} y_s A_s \right)$
 where $y_s = A_s^T \theta + \eta_s$

We have
$$\begin{aligned} \hat{\theta}_t &= V_t^{-1} \left(\sum_{s=1}^{t-1} (A_s^T \theta + \eta_s) A_s \right) \\ &= V_t^{-1} \left(\underbrace{\left(\sum_{s=1}^{t-1} A_s A_s^T \right)}_{\approx V_t} \theta + \underbrace{\sum_{s=1}^{t-1} A_s \eta_s}_{\text{"small" because } \eta_s \sim \mathcal{N}(0, \sigma^2)} \right) \\ &\approx \theta + V_t^{-1} \left(\sum_s A_s \eta_s \right) \end{aligned}$$

So $V_t^{\frac{1}{2}} (\hat{\theta}_t - \theta) \approx V_t^{-\frac{1}{2}} p_t$

$$\left(\hat{\theta}_t - \theta \right)^T V_t^{\frac{1}{2}} V_t^{\frac{1}{2}} (\hat{\theta}_t - \theta) = p_t^T V_t^{-1} p_t$$

$$\|\hat{\theta}_t - \theta\|_{V_t}^2 \approx \|p_t\|_{V_t^{-1}}^2$$

$$\|X\|_M = \sqrt{X^T M X}$$

Confidence Ellipsoids

What's in $V_t(\omega)$? $V_t(\omega) = \sum_s A_s A_s^\top + \lambda I \rightarrow$

So $\|\hat{\theta}_t - \theta\|_{V_t}^2 \approx \|p_t\|_{V_t^{-1}}^2$. Define

$$\mathcal{C}_t(\delta) = \left\{ \theta \in \mathcal{B}_{1,d} : \|\theta - \hat{\theta}_t\|_{V_t} \leq \beta_t \right\}$$

Theorem: Let $\beta_t = \sqrt{\lambda} + \sqrt{2 \log\left(\frac{1}{\delta}\right) + d \log\left(\frac{\lambda d + t}{\lambda d}\right)}$

Then, w.p. at least $1 - \delta$,

$$\theta \in \mathcal{C}_t(\delta)$$

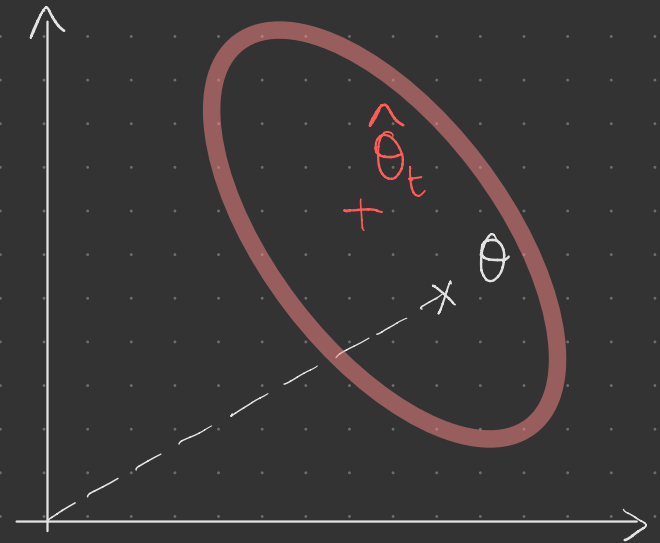
proof: beyond the scope of this class (see chapters 19-21 of Bandit Algorithms) but it's a major result of the field.

Important remark: Recent advances tighten these bounds: see (Flynn et al. 2023)

$d = 2$

$$\begin{pmatrix} \sum_s a_{s(1)} a_{s(1)} & \sum_s a_{s(2)} a_{s(1)} \\ \sum_s a_{s(1)} a_{s(2)} & \sum_s a_{s(2)} a_{s(2)} \end{pmatrix} \begin{matrix} = N_1(t) \\ = N_2(t) \end{matrix}$$

if $A \in \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$



LinUCB

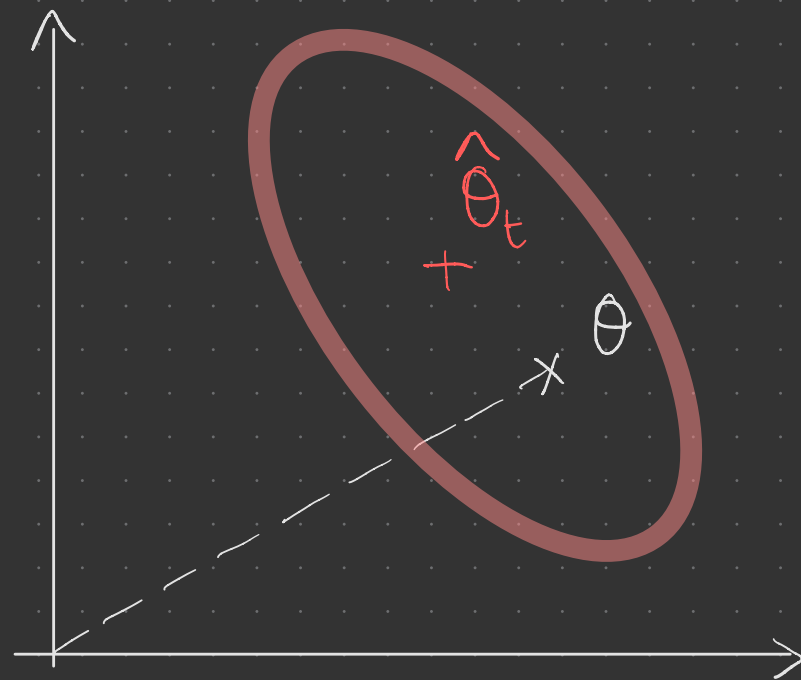
Optimism principle:

"pick the arm with the best plausible reward in all plausible environments (with high probability)"

$$A_t = \operatorname{argmax}_{a \in \mathcal{A}_t} \underbrace{\max_{\tilde{\theta} \in \mathcal{B}_t(t)} a^\top \tilde{\theta}}_{\text{UCB}(a)}$$

Note: $\mathcal{A}_t$ is finite (no computational issue for the first max)

Computing the second maximum?



To compute the UCB for arm a , we need to solve the following optimization problem:

$$\begin{aligned} \max_{\theta} \quad & \theta^T a \\ \text{s.t.} \quad & \|\theta - \hat{\theta}_t^{\text{RLS}}\|_{V_t(x)} \leq \beta_t \end{aligned} \quad \Leftrightarrow \quad \begin{aligned} \max_{\Delta} \quad & \hat{\theta}_t^T a + \Delta^T a \\ \text{s.t.} \quad & \|\Delta\|_{V_t(x)}^2 \leq \beta_t^2 \end{aligned}$$

→ maximize linear function under convex constraints! OK ✓

exercise: Solve this linear program

↳ The solution yields ...

Recall that $\mathcal{P}_t(S)$ has "diameter" $\beta_t \rightarrow$ high proba coverage

Lin UCB: at round t , choose arm $A_t \in \arg\max_{a \in \mathcal{I}_t} \hat{\theta}_t^T a + \beta_t \|a\|_{V_t(x)}^{-1}$

$$\max_{\Delta, t} \quad \hat{\theta}^T a + \Delta^T a = -\min_{\Delta} -\hat{\theta}^T a - \Delta^T a$$

$$s.t. \quad \|\Delta\|_{V_t(\lambda)}^2 \leq \beta_t^2 \Leftrightarrow \Delta^T V_t(\lambda) \Delta - \beta_t^2 \leq 0$$

$$\mathcal{L}(\Delta, \nu) = -\hat{\theta}^T a - \Delta^T a + \nu (\Delta^T V_t(\lambda) \Delta - \beta_t^2)$$

$$\nabla \mathcal{L}(\Delta; \nu) = -a + 2\nu V_t(\lambda) \Delta$$

$$\Delta^* = \frac{1}{2\nu} V_t(\lambda)^{-1} a \quad \text{with } \|\Delta^*\|_{V_t(\lambda)} = \frac{1}{4\nu^2} \|a\|_{V_t(\lambda)}^2$$

$$g(\nu) = \inf_{\Delta} \mathcal{L}(\Delta; \nu) = -\hat{\theta}^T a - \frac{1}{2\nu} a^T V_t(\lambda)^{-1} a + \nu \left(\frac{\|a\|_{V_t(\lambda)}^2}{4\nu^2} - \beta_t^2 \right)$$

$$= -\hat{\theta}^T a - \frac{1}{4\nu} \|a\|_{V_t(\lambda)}^2 - \nu \beta_t^2$$

$$\nu^* = \max_{\nu \geq 0} g(\nu) =$$

$$g'(\nu) = + \frac{1}{4\nu^2} \|a\|_{V_t(\lambda)}^2 - \beta_t^2$$

$$\rightarrow \frac{1}{\nu^{*2}} = \frac{4\beta_t^2}{\|a\|_{V_t(\lambda)}^2} \quad s.t. \quad \frac{\|a\|_{V_t(\lambda)}^2}{2\beta_t} = \nu^* \quad (\nu \geq 0)$$

$$\text{whence } \Delta^* = \frac{1}{2\nu^*} V_t(\lambda)^{-1} a = \frac{\beta_t}{\|a\|_{V_t(\lambda)}^2} V_t(\lambda)^{-1} a$$

$$\text{and the desired maximum is } \boxed{\hat{\theta}^T a + \beta_t \|a\|_{V_t(\lambda)}^{-2}}$$

⊕ Simple form of the solution: greedy + bonus

⊖ Need to compute $\|a\|_{V_t(\lambda)}^{-2} = a^T V_t(\lambda)^{-1} a$

Regret(s)?

$$a_t^{*\top} \theta$$

First recall that: $R(T) = \sum_{t=1}^T \overbrace{\max_{a \in \mathcal{A}_t} a^\top \theta}^{a_t^{*\top} \theta} - A_t^\top \theta$

Regret of LinUCB

$$a_t^{*\top} \theta \leq \text{UCB}(a_t^*) \quad (\text{w.p. } \geq 1-\delta, \text{ by def of UCB})$$

$$\leq \text{UCB}(A_t) \quad (\text{by design of LinUCB})$$

$$= A_t^\top \tilde{\theta}$$

$$\leftarrow \exists \tilde{\theta} \in \mathcal{B}_t(\delta) \text{ st. } \text{UCB}(A_t) = \tilde{\theta}^\top A_t$$

$$\hookrightarrow a_t^{*\top} \theta - A_t^\top \theta \leq A_t^\top (\theta - \tilde{\theta})$$

not too big because $\tilde{\theta} \in \mathcal{B}_t(\delta)$ 

$$\leq \|A_t\|_{V_t^{-1}} \underbrace{\|\theta - \tilde{\theta}\|_{V_t}}_{\leq 2\beta_t}$$

Theorem: The regret of LinUCB (bounded actions and rewards) is bounded by $R(T) = \mathcal{O}(d \sqrt{T} \log(T))$

proof: $R(T) \leq \sqrt{T \sum \|A_t\|_{V_t^{-1}}^2} \times 4\beta_t \leq \sqrt{T \underbrace{\beta_T^2}_d \underbrace{\sum_t \|A_t\|_{V_t^{-1}}^2}_{\leq d}} \quad \text{Elliptical lemma (Chap 19)}$

Regret(s)?

How hard is this regret minimization problem in general?

Theorem: If $\mathcal{A} = \mathcal{S}_{1,d}$ and $\theta \in \left\{-\frac{1}{\sqrt{T}}, \frac{1}{\sqrt{T}}\right\}^d$
 $R(T) = \Omega(d\sqrt{T})$ ✓ (Chap 24)

LinUCB is minimax optimal (up to log terms)

Is that all?

* Fixed-design setting: $\mathcal{A} = \{e_1, e_2, \tilde{e}\}$

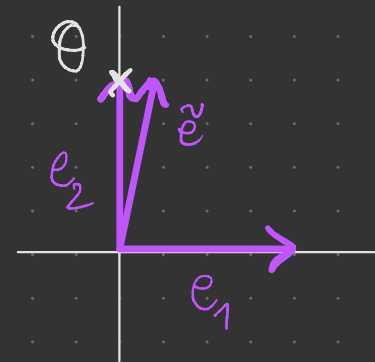
→ The end of optimism? (Lattimore et al. 2017)

Chap. 21, 22

* Algorithms for both settings? e.g. (Kirschner et al. 2020)

<https://arxiv.org/abs/2011.05944> (IDS)

* Any other setting?



LinTS: The Bayesian Way

The idea of Thompson Sampling is to maintain a posterior distribution on θ from which the learner samples at each round to make his decision.

Let us denote π_0 the prior on θ . For instance, $\pi_0 \sim \mathcal{N}(0, \lambda I_d)$ for some $\lambda > 0$.

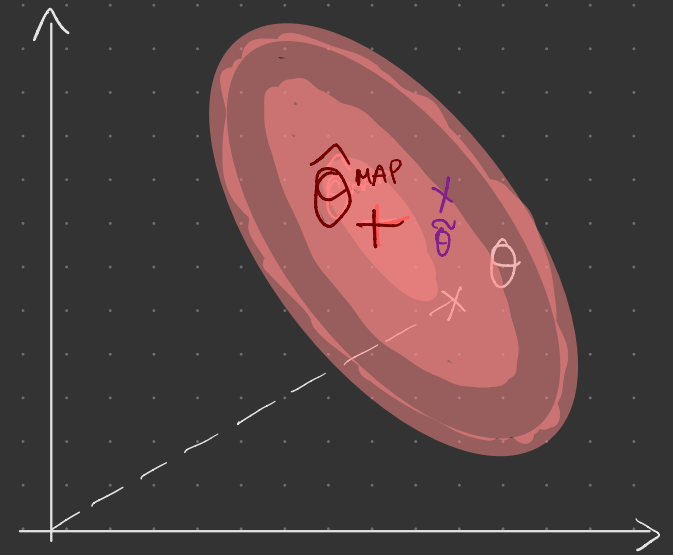
We assume that the reward noise is Gaussian:
$$Y_t = A_t^\top \theta + \eta_t \quad \text{where } \eta_t \sim \mathcal{N}(0, \sigma^2)$$

By Bayes' Theorem, the posterior after 1 observation $A_1 Y_1$

$$\begin{aligned} \text{is } \pi_1(\theta; A_1, Y_1) &= \frac{p(Y_1 | \theta, A_1) \times \pi_0(\theta)}{\int p(Y_1 | \theta, A_1) \times \pi_0(\theta) d\theta} \\ &\propto \underbrace{A_1^\top Y_1}_{\downarrow \pi_1(\theta)} \propto \mathcal{N}(\theta^\top A_1, \sigma^2) \mathcal{N}(0, \lambda I) \end{aligned}$$

↳ product of Gaussians so it's a Gaussian

Exercise: Derive the posterior distribution at time t , π_t , conditional on the actions $A_1, \dots, A_{t-1}$ and the known variance of the rewards σ^2 .



LinTS:

* Sample $\tilde{\theta} \sim \pi_t$ from current posterior

* Choose

$$A_t \in \operatorname{argmax}_{a \in \mathcal{A}_t} \tilde{\theta}^\top a$$

II. Beyond Linear Models

1 - Kernelized bandits and GP-UCB

1.5 - NTK bandits (Kassraie & Krause 2022)

2 - Deep Bayesian Bandits Showdown

3 - Beyond UCB